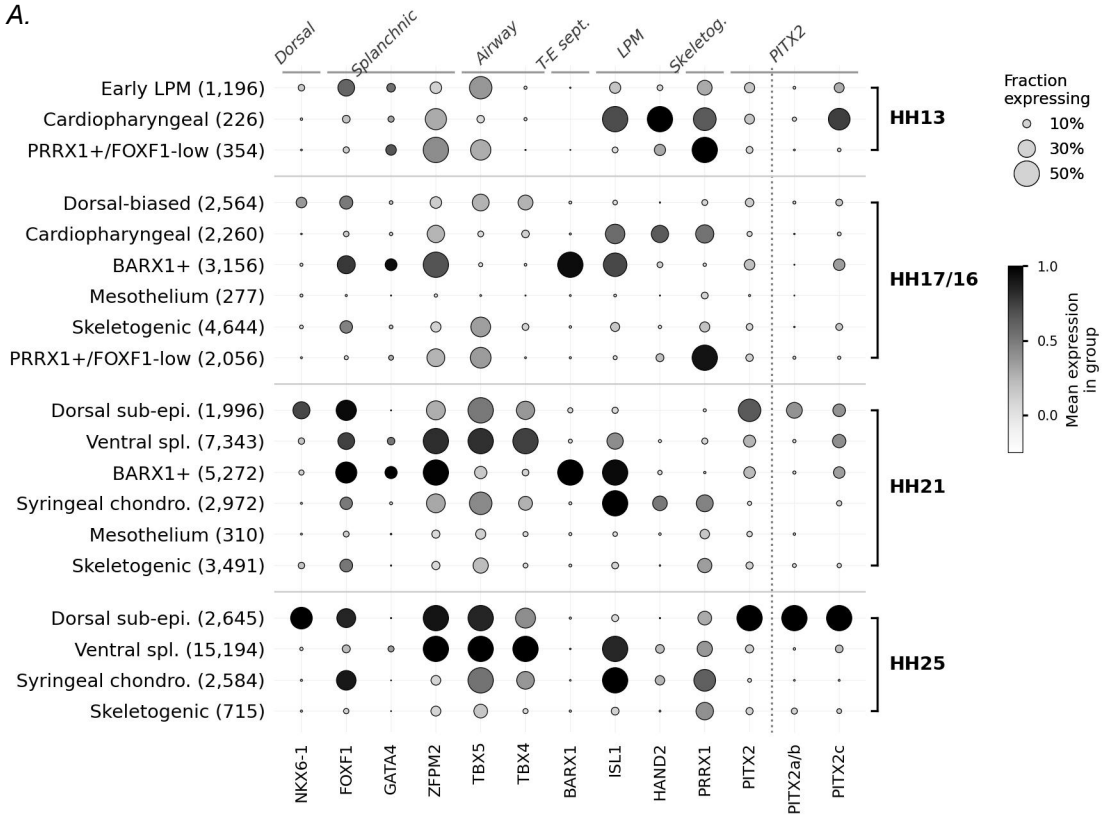
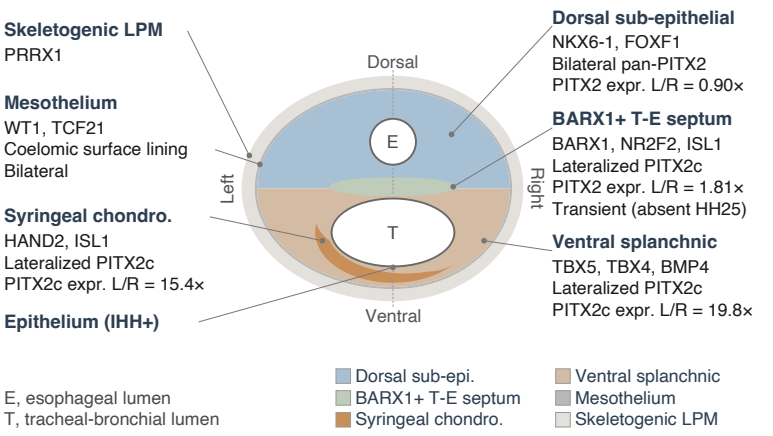
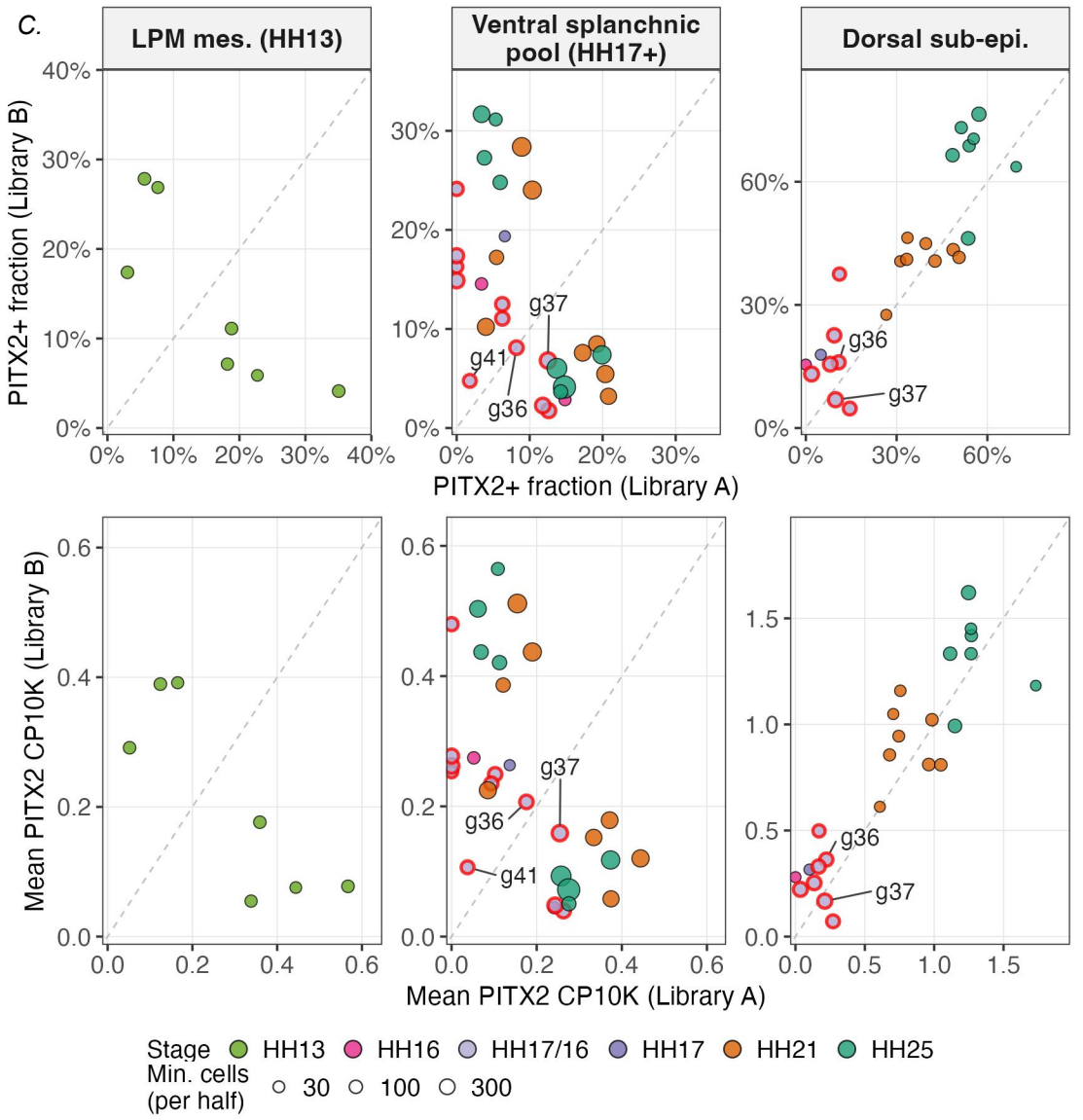
A.


Dorsal
Splanchnic
Airway
T-E sept.
LPM
Skeletog.
PITX2
Early LPM (1,196)
Cardiopharyngeal (226)
PRRX1+/FOXF1-low (354)
Dorsal-biased (2,564)
Cardiopharyngeal (2,260)
BARX1+ (3,156)
Mesothelium (277)
Skeletogenic (4,644)
PRRX1+/FOXF1-low (2,056)
Dorsal sub-epi. (1,996)
Ventral spl. (7,343)
BARX1+ (5,272)
Syringeal chondro. (2,972)
Mesothelium (310)
Skeletogenic (3,491)
Dorsal sub-epi. (2,645)
Ventral spl. (15,194)
Syringeal chondro. (2,584)
Skeletogenic (715)
HH13
HH17/16
HH21
HH25
NKX6-1
FOXF1
GATA4
ZFPM2
TBX5
TBX4
BARX1
ISL1
HAND2
PRRX1
PITX2
PITX2a/b
PITX2c
Fraction expressing
10%
30%
50%
Mean expression in group
1.0
0.5
0.0

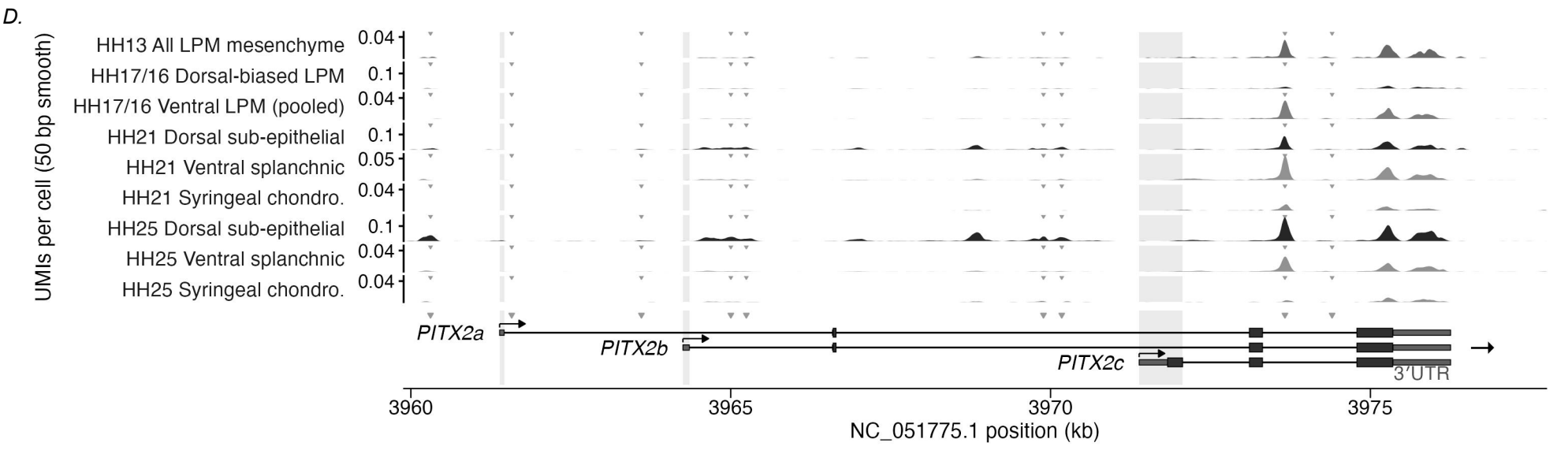
B.


Skeletogenic LPM
PRRX1
Mesothelium
WT1, TCF21
Coelomic surface lining
Bilateral
Syringeal chondro.
HAND2, ISL1
Lateralized PITX2c
PITX2c expr. L/R = 15.4×
Epithelium (IHH+)
Dorsal sub-epithelial
NKX6-1, FOXF1
Bilateral pan-PITX2
PITX2 expr. L/R = 0.90×
BARX1+ T-E septum
BARX1, NR2F2, ISL1
Lateralized PITX2c
PITX2 expr. L/R = 1.81×
Transient (absent HH25)
Ventral splanchnic
TBX5, TBX4, BMP4
Lateralized PITX2c
PITX2c expr. L/R = 19.8×
Dorsal
Ventral
Left
Right
E
T
E, esophageal lumen
T, tracheal-bronchial lumen
Dorsal sub-epi.
BARX1+ T-E septum
Syringeal chondro.
Ventral splanchnic
Mesothelium
Skeletogenic LPM

C.


LPM mes. (HH13)
Ventral splanchnic pool (HH17+)
Dorsal sub-epi.
PITX2+ fraction (Library B)
PITX2+ fraction (Library A)
Mean PITX2 CP10K (Library B)
Mean PITX2 CP10K (Library A)
g37
g36
g41
Stage
HH13
HH16
HH17/16
HH17
HH21
HH25
Min. cells (per half)
30
100
300

D.


UMIs per cell (50 bp smooth)
HH13 All LPM mesenchyme
HH17/16 Dorsal-biased LPM
HH17/16 Ventral LPM (pooled)
HH21 Dorsal sub-epithelial
HH21 Ventral splanchnic
HH21 Syringeal chondro.
HH25 Dorsal sub-epithelial
HH25 Ventral splanchnic
HH25 Syringeal chondro.
PITX2a
PITX2b
PITX2c
3′UTR
NC_051775.1 position (kb)